Sharp Prime–Schatten Thresholds for Finite-Weil Rauzy–AGY Transfer Operators

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Abstract

For each odd prime p, a source-locked Rauzy–AGY transfer operator twisted by the full finite Weil representation acts on a Bergman space with a p^2-dimensional fibre. We determine exactly whether these local operators assemble into a prime-independent global determinant. Under the proved nuclear branch majorant and fixed-start matrix-decoder hypotheses, we show, locally uniformly in the dynamical parameter, that

$$\|\mathcal{L}_{s,p}\|_{S_q} \asymp p^{2/q} \qquad (1 \le q < \infty),$$

with a prime-uniform positive operator-norm lower bound. Consequently the block sum $\bigoplus_p c_p \mathcal{L}_{s,p}$ belongs to S_q if and only if $\sum_p p^2 |c_p|^q < \infty$. For $c_p = p^{-z}$, this gives the sharp wall $q \operatorname{Re} z > 3$, and an ordinary, prime-order-independent Fredholm product precisely for $\operatorname{Re} z > 3$. The unweighted sum is noncompact. In the other weight-free normalization, finite-Weil characters converge pointwise to the regular character; freeness of the positive return monoid then forces $\exp[p^{-2} \operatorname{Log}_0 \mathcal{D}_p(s,u)] \to 1$ on a common small disc. We also identify orbitwise quadratic prime series at regular words and give a strictly scoped fixed-plane control from the full Rauzy ledger. Thus a global determinant exists after adding an external prime clock, but the two canonical undamped constructions fail in complementary ways. No adelic, automorphic, or Hilbert–Pólya conclusion is asserted.

1 Introduction

A determinant attached to one finite prime is local data. A canonical global object should additionally explain how the prime fibres are assembled, why the assembly is independent of an ordering of the primes, and whether the normalization is intrinsic to the original dynamical clock. This distinction is particularly sharp for transfer operators twisted by finite Weil representations: the chronological orbit product is retained exactly at every prime, but the fibre dimension grows as p^2.

This paper answers the resulting assembly question for a fixed Rauzy–AGY symbolic clock. The scalar operators arise from an accelerated return map of the kind used by Avila, Gouëzel and Yoccoz in their study of Teichmüller dynamics [2]. Every return branch carries an integral symplectic matrix. Reduction modulo an odd prime and the full finite Weil representation produce

$$\mathcal{L}_{s,p} = \sum_{\gamma \in \Gamma} K_{s,\gamma} \otimes \rho_p(g_\gamma) \quad \text{on} \quad \mathcal{H}_p = A^2(\Omega) \widehat{\otimes} \mathbb{C}^{p^2}. \tag{1}$$

The sum is chronological: a word $w = (\gamma_1, \ldots, \gamma_n)$ contributes the actual ordered product g_w, with later forward returns multiplying on the left. We never replace this product by an average of transition matrices, a product of characters, or a commutative surrogate.

1.1 Main results

Our first result shows that the evident p^2 trace-norm upper bound is sharp in every Schatten class. For every compact parameter set and every $1 \le q < \infty$,

$$c_{q,K} p^{2/q} \le \|\mathcal{L}_{s,p}\|_{S_q} \le C_{q,K} p^{2/q} \tag{2}$$

for all sufficiently large odd primes and all $s \in K$. The lower bound is not a dimension count. Constant functions and evaluation at one interior point compress $\mathcal{L}_{s,p}$ to an absolutely summable linear combination of Weil matrices. Pairing that compression with the inverse of one branch matrix isolates a nonzero coefficient. Two ingredients make the isolation uniform: the fixed-start full-matrix decoder rules out an off-diagonal identity, and the normalized Weil characters of each fixed nonidentity matrix tend to zero.

The sharp estimate yields the exact block-sum classification

$$\bigoplus_{p \text{ odd}} c_p \mathcal{L}_{s,p} \in S_q \quad \Longleftrightarrow \quad \sum_{p \text{ odd}} p^2 |c_p|^q < \infty. \tag{3}$$

In particular, for $c_p = p^{-z}$,

$$\mathfrak{L}_{s,z} := \bigoplus_{p \text{ odd}} p^{-z} \mathcal{L}_{s,p} \in S_q \quad \Longleftrightarrow \quad q \operatorname{Re} z > 3. \tag{4}$$

The equality line is excluded by divergence of the prime harmonic series. Thus the ordinary Fredholm determinant

$$\mathfrak{D}(s,z,u) = \det(I - u\mathfrak{L}_{s,z}) = \prod_{p \text{ odd}} \mathcal{D}_p(s, u p^{-z}) \tag{5}$$

exists for $\operatorname{Re} z > 3$, is jointly holomorphic, and is independent of the enumeration of the primes. The prime factor p^{-z} does not erase chronological dynamics, but it adds the external per-return clock $z \log p$.

The two undamped alternatives fail differently. The counting-trace direct sum is noncompact because the local operator norms stay uniformly away from zero. If instead the fibre trace is normalized by p^{-2}, the finite Weil characters converge pointwise to the regular character of the integral cocycle group. The positive return monoid is free, so no nonempty word is an identity. Every normalized positive moment vanishes and, on a common small disc,

$$\exp\bigl[p^{-2} \operatorname{Log}_0 \mathcal{D}_p(s,u)\bigr] \longrightarrow 1. \tag{6}$$

Here Log_0 is the Fredholm-logarithm branch fixed by value zero at $u = 0$; equation (6) is not a global choice of a p^2-th root.

1.2 Relation to existing structures

Three established theories meet in the construction but do not identify the global object automatically. Rauzy–AGY acceleration supplies the countable chronological grammar and its nuclear scalar transfer operator [2]. Finite-field quantization supplies a canonical Weil representation and exact character formulae [4, 6]. Trace-ideal theory supplies Fredholm and regularized determinants once the correct Schatten threshold is known [3, 5]. The new point is the sharp lower bound linking these three inputs.

The prime direct sum in equation (5) is not an adelic Weil representation. The latter belongs to a restricted tensor product of local-field oscillator representations with compatible global additive characters, measures, splittings, and almost-everywhere reference vectors [7]. Residue-field spaces indexed by primes and joined by a Hilbert direct sum do not provide those data.

1.3 Scope

The conclusions are operator-theoretic and arithmetic at the level of orbitwise prime coefficients. We prove no continuation to $z = 0$, common automorphic conductor, functional equation, Riemann–von Mangoldt law, self-adjoint Hilbert–Pólya operator, or relation between the zero set of $\mathfrak{D}$ and the Riemann zeros. A fixed-plane example used later is a *marked full-Rauzy control*; it is never promoted to a branch of the narrower induced language.

The source-locked hypotheses are stated in section 2. The normalized character limit and sharp Schatten theorem occupy sections 3 and 4. The determinant and normalized-trace limits appear in sections 5 and 6. Orbitwise arithmetic and the fixed-plane control are separated in section 7; interpretation and reproducibility follow in sections 8 and 9 and appendix A.

2 Source-locked dynamical and analytic setting

We isolate the precise inherited inputs so that every new implication can be checked without appealing to a finite prime window. Let

$$\mathfrak{S} = \{s \in \mathbb{C} : \operatorname{Re} s > -\sigma_0\}, \qquad \mathcal{H}_0 = A^2(\Omega),$$

where $\Omega \Subset \mathbb{C}^3$ is a bounded domain. The countable alphabet Γ consists of first-return branches based at one labeled Rauzy state. The scalar branch operator is the holomorphic weighted composition operator $K_{s,\gamma}$ on $\mathcal{H}_0$.

Hypothesis 2.1 (Nuclear branch majorant)**.** *For every compact $K \Subset \mathfrak{S}$,*

$$C_{1,K} := \sup_{s\in K} \sum_{\gamma\in\Gamma} \|K_{s,\gamma}\|_{S_1} < \infty. \tag{7}$$

The sum has locally uniform tails: for every $\varepsilon > 0$ there is a finite $F \subset \Gamma$ such that

$$\sup_{s\in K} \sum_{\gamma\notin F} \|K_{s,\gamma}\|_{S_1} < \varepsilon. \tag{8}$$

The branches depend holomorphically on s in trace norm.

Hypothesis 2.2 (A nonvanishing scalar slice)**.** *There is a real interior point $x_0 \in \Omega$ such that constant embedding $\iota_0 : \mathbb{C} \to \mathcal{H}_0$ and evaluation $E_0 : \mathcal{H}_0 \to \mathbb{C}$ are bounded. Put*

$$a_\gamma(s) = E_0 K_{s,\gamma} \iota_0. \tag{9}$$

Every a_γ is holomorphic and nowhere zero on $\mathfrak{S}$, and for compact $K \Subset \mathfrak{S}$,

$$\sup_{s\in K} \sum_{\gamma\in\Gamma} |a_\gamma(s)| < \infty. \tag{10}$$

For every $\varepsilon > 0$, after enlarging a finite $F \subset \Gamma$ if necessary,

$$\sup_{s\in K} \sum_{\gamma\notin F} |a_\gamma(s)| < \varepsilon. \tag{11}$$

Hypothesis 2.3 (Fixed-start full-matrix decoder)**.** *Every branch carries a matrix $g_\gamma \in \mathrm{Sp}(J_0, \mathbb{Z})$, where J_0 is the fixed integral symplectic form of rank four. For the frozen starting labeled permutation, the full nonnegative Rauzy matrix and the start state recover the complete directed edge word. Distinct first returns therefore have distinct matrices. Moreover, concatenations split uniquely at visits to the base state, so the positive first-return matrix monoid is free.*

This is the exact content used from the predecessor decoder theorem (C25, Theorem E.1 and Corollary E.2). Its peeling algorithm detects the unique final winner–loser pair by row domination, subtracts the loser row from the winner row, and strictly decreases the total matrix-entry sum. Iteration reaches the identity and recovers the labeled edge word. Unique splitting at base-state returns then gives

$$\gamma \neq \delta \Longrightarrow g_\gamma g_\delta^{-1} \neq I, \qquad w \neq \varnothing \Longrightarrow g_w \neq I. \tag{12}$$

The statement concerns full matrices, not characteristic polynomials, spectra, conjugacy classes, or characters. It is also an all-length theorem, not a collision search up to a cutoff.

Hypothesis 2.4 (Full finite-Weil fibres)**.** *For every odd prime p, let ρ_p be the full genuine finite Weil representation of the rank-four symplectic space in the source frame. It is unitary on $\mathbb{C}^{p^2}$. For*

$$\Theta_p(h) = \operatorname{Tr} \rho_p(h \bmod p), \tag{13}$$

the character magnitude is

$$|\Theta_p(h)|^2 = |\ker_{\mathbb{F}_p}(h - I)|. \tag{14}$$

Equation (14) is the specialization of the exact Weil character formula [6]; finite-field constructions and normalizations are discussed in Gurevich and Hadani [4]. We use the full p^2-dimensional representation, not an irreducible parity summand.

Under hypotheses 2.1 and 2.4, the local operator

$$\mathcal{L}_{s,p} = \sum_{\gamma \in \Gamma} K_{s,\gamma} \otimes \rho_p(g_\gamma) \tag{15}$$

converges in trace norm on $\mathcal{H}_p = \mathcal{H}_0 \widehat{\otimes} \mathbb{C}^{p^2}$ and is holomorphic in s. Its ordinary local determinant is

$$\mathcal{D}_p(s, u) = \det_{\mathcal{H}_p}(I - u\mathcal{L}_{s,p}). \tag{16}$$

2.1 Chronology convention

If $w = (\gamma_1, \ldots, \gamma_n)$, the operator product in $\mathcal{L}_{s,p}^n$ fixes the ordered cocycle product g_w. Later forward returns multiply on the left, according to the frozen convention. The scalar fixed-point trace atom is denoted

$$A_{s,w} = \frac{\lambda_w^{-(s+1)}}{\chi_w'(\lambda_w)}. \tag{17}$$

The inherited absolute word trace formula is

$$\operatorname{Tr} \mathcal{L}_{s,p}^n = \sum_{w \in \Gamma^n} A_{s,w} \Theta_p(g_w), \tag{18}$$

with a locally uniform absolute majorant after division by p^2. Explicitly, for every fixed n, compact $K \Subset \mathfrak{S}$, and $\varepsilon > 0$, there is a finite $W \subset \Gamma^n$ such that

$$\sup_{s \in K} \sum_{w \in \Gamma^n \setminus W} |A_{s,w}| < \varepsilon. \tag{19}$$

A primitive word repeated r times has fibre factor $\Theta_p(g_w^r)$, never $\Theta_p(g_w)^r$. These conventions preserve genuine noncommutative chronology throughout the prime assembly.

3 The normalized finite-Weil character limit

The large-prime mechanism is elementary once the exact character magnitude is retained. Its pointwise nature is important.

Lemma 3.1 (Eventual rank and character magnitude)**.** *Fix $h \in \mathrm{Sp}(J_0, \mathbb{Z})$ and set*

$$r(h) = \mathrm{rank}_{\mathbb{Q}}(h - I).$$

Outside a finite set of odd primes,

$$\mathrm{rank}_{\mathbb{F}_p}(h - I) = r(h), \qquad \frac{|\Theta_p(h)|}{p^2} = p^{-r(h)/2}. \tag{20}$$

Proof. Every minor of $h - I$ of size greater than $r(h)$ vanishes as an integer. At least one $r(h)$-minor is a nonzero integer d. If $p \nmid d$, that minor remains nonzero modulo p, whereas the larger minors still vanish. Thus the rank over $\mathbb{F}_p$ is exactly $r(h)$ outside the finite set of prime divisors of d. The fixed-space dimension is then $4 - r(h)$. By equation (14),

$$|\Theta_p(h)|^2 = p^{4 - r(h)},$$

and division by p^4, followed by a square root, gives equation (20). □

Corollary 3.2 (Pointwise regular-character limit)**.** *For every fixed integral cocycle element h,*

$$\frac{\Theta_p(h)}{p^2} \longrightarrow \delta_I(h) = \begin{cases} 1, & h = I, \\ 0, & h \neq I. \end{cases} \tag{21}$$

Proof. For $h = I$, the character equals the representation dimension p^2. For $h \neq I$, one has $r(h) \geq 1$, and lemma 3.1 makes the modulus of the normalized character tend to zero. □

Normalized characters of finite-dimensional unitary representations are positive definite. Thus δ_I is naturally the regular character of the discrete subgroup generated by the integral cocycle matrices. The corollary is fully source-locked for each *fixed* integral matrix. It does not assert uniform convergence over a word set whose length or matrix height grows with p. Later countable-sum applications first use the pointwise limit on a finite head and then control the complement by the locally uniform tail hypotheses; they never assume a global uniform character limit.

4 Sharp local Schatten growth

We now prove the estimate that controls every global construction. Schatten norms and their duality are used in the standard sense of Simon [5].

Theorem 4.1 (Sharp finite-Weil multiplicity)**.** *Let $K \Subset \mathfrak{S}$ be compact. For every $1 \le q < \infty$, there are constants $0 < c_{q,K} \le C_{q,K} < \infty$ and a prime threshold p_K such that*

$$c_{q,K} p^{2/q} \le \|\mathcal{L}_{s,p}\|_{S_q} \le C_{q,K} p^{2/q} \tag{22}$$

for every odd $p \ge p_K$ and every $s \in K$. There are likewise constants, uniform on K, such that

$$0 < c_{\infty,K} \le \|\mathcal{L}_{s,p}\| \le C_{\infty,K}. \tag{23}$$

Proof. We separate the upper estimate from the branch-isolating lower estimate.

Upper bound. If T is compact and U is unitary on an N-dimensional space, the singular values of $T \otimes U$ are those of T, each repeated N times. Since $N = p^2$,

$$\|K_{s,\gamma} \otimes \rho_p(g_\gamma)\|_{S_q} = p^{2/q} \|K_{s,\gamma}\|_{S_q}. \tag{24}$$

The triangle inequality, $\|T\|_{S_q} \le \|T\|_{S_1}$, and equation (7) imply

$$\|\mathcal{L}_{s,p}\|_{S_q} \le p^{2/q} \sum_{\gamma} \|K_{s,\gamma}\|_{S_q} \le C_{1,K} p^{2/q}.$$

The same calculation in operator norm, without the multiplicity factor, gives the upper bound in equation (23).

Compression and locally uniform isolation. Tensor E_0 and ι_0 with the identity on $\mathbb{C}^{p^2}$, obtaining E_p and ι_p. Their norms do not depend on p. By equation (9),

$$B_{s,p} := E_p \mathcal{L}_{s,p} \iota_p = \sum_{\gamma \in \Gamma} a_\gamma(s) \rho_p(g_\gamma). \tag{25}$$

Fix one branch $\delta \in \Gamma$ once and for all, and put $\tau_p = p^{-2}\,\mathrm{Tr}$. The exact normalized pairing is

$$\tau_p\big(B_{s,p}\rho_p(g_\delta)^{-1}\big) = a_\delta(s) + R_p(s), \qquad R_p(s) = \sum_{\gamma \ne \delta} a_\gamma(s) \frac{\Theta_p(g_\gamma g_\delta^{-1})}{p^2}. \tag{26}$$

Every matrix in the remainder is nonidentity by equation (12), so each normalized character tends to zero by corollary 3.2. Moreover,

$$\big|p^{-2}\Theta_p(h)\big| \le 1$$

because it is the normalized trace of a unitary. We claim that $R_p \to 0$ uniformly on K. Given $\varepsilon > 0$, choose a finite set $F \subset \Gamma$, containing δ, for which

$$\sup_{s \in K} \sum_{\gamma \notin F} |a_\gamma(s)| < \varepsilon;$$

this is possible by equation (10). The tail of R_p is then at most ε, uniformly in s and p. On the finite head $F \setminus \{\delta\}$, every normalized character tends to zero and the coefficient sum is bounded on K. Hence the head is uniformly small for all sufficiently large p. This finite-head/uniform-tail argument proves

$$\tau_p\bigl(B_{s,p}\rho_p(g_\delta)^{-1}\bigr) \longrightarrow a_\delta(s) \quad \text{uniformly for } s \in K. \tag{27}$$

Since a_δ is continuous and nowhere zero,

$$m_{\delta,K} := \min_{s\in K} |a_\delta(s)| > 0. \tag{28}$$

Consequently the modulus of the normalized pairing in equation (27) is at least $m_{\delta,K}/2$, simultaneously for $s \in K$, once p is large enough.

Schatten duality. Let q' be conjugate to q, with $q' = \infty$ when $q = 1$. The trace pairing and the unitary singular values give

$$\frac{m_{\delta,K}}{2} p^2 \le \bigl|\mathrm{Tr}\bigl(B_{s,p}\rho_p(g_\delta)^{-1}\bigr)\bigr| \tag{29}$$

$$\le \|B_{s,p}\|_{S_q} \bigl\|\rho_p(g_\delta)^{-1}\bigr\|_{S_{q'}} = \|B_{s,p}\|_{S_q} p^{2/q'}. \tag{30}$$

Thus $\|B_{s,p}\|_{S_q} \ge (m_{\delta,K}/2)p^{2/q}$. The ideal inequality

$$\|B_{s,p}\|_{S_q} \le \|E_p\| \; \|\mathcal{L}_{s,p}\|_{S_q} \; \|\iota_p\|$$

transfers this lower bound to $\mathcal{L}_{s,p}$. For the operator norm, use the trace pairing with the trace norm of the unitary, equal to p^2, and apply the same ideal inequality. This proves both statements. □

The proof uses the all-length matrix decoder, not injectivity of a coarser spectral signature. It also controls the entire countable induced alphabet: the summable tail is removed before the prime limit is applied.

4.1 Exact block-sum classification

Let $c = (c_p)_{p \text{ odd}}$ be complex scalars. On the algebraic direct sum inside

$$\mathcal{H}_\oplus = \bigoplus_{p \text{ odd}} \mathcal{H}_p,$$

consider the block-diagonal family

$$\mathbb{L}_s(c) = \bigoplus_{p \text{ odd}} c_p \mathcal{L}_{s,p}. \tag{31}$$

Theorem 4.2 (Prime–Schatten criterion)**.** *For every fixed $s \in \mathfrak{S}$ and $1 \le q < \infty$,*

$$\mathbb{L}_s(c) \in S_q \quad \Longleftrightarrow \quad \sum_{p \text{ odd}} p^2 |c_p|^q < \infty. \tag{32}$$

The block family extends to a compact operator if and only if $c_p \to 0$. Whenever equation (32) *holds, its convergence in S_q is locally uniform in s.*

Proof. For a block-diagonal compact operator, the singular-value multiset is the disjoint union of the block singular-value multisets. Hence

$$\|\mathbb{L}_s(c)\|_{S_q}^q = \sum_{p \text{ odd}} |c_p|^q \, \|\mathcal{L}_{s,p}\|_{S_q}^q . \tag{33}$$

The two sides of equation (22), outside finitely many primes, compare the summand in equation (33) with $p^2|c_p|^q$. This proves the equivalence. The upper constant in theorem 4.1, uniform on compact s-sets, also makes the tails locally uniform.

Every local block is compact. A block-diagonal family of compact operators is compact precisely when its block norms tend to zero. By equation (23), this condition is equivalent to $c_p \to 0$. □

Corollary 4.3 (Sharp Dirichlet phase diagram)**.** *For*

$$\mathfrak{L}_{s,z} = \bigoplus_{p \text{ odd}} p^{-z}\mathcal{L}_{s,p}, \qquad p^{-z} := e^{-z\log p}, \tag{34}$$

one has

$$\mathfrak{L}_{s,z} \in S_q \quad \iff \quad q \operatorname{Re} z > 3, \qquad 1 \le q < \infty. \tag{35}$$

It is compact exactly when $\operatorname{Re} z > 0$*.*

Proof. Write $\sigma = \operatorname{Re} z$. The series in equation (32) becomes

$$\sum_{p \text{ odd}} p^{2-q\sigma}.$$

The prime series $\sum_p p^{-\beta}$ converges exactly for $\beta > 1$, and diverges at $\beta = 1$ [1]. Thus $q\sigma > 3$ is necessary and sufficient, including strict exclusion of the boundary. Compactness follows from $|p^{-z}| = p^{-\sigma}$ and theorem 4.2. □

5 The prime-graded determinant

Trace class is the exact region in which the block assembly gives an ordinary Fredholm determinant without choosing an order of multiplication.

Theorem 5.1 (Prime-order-independent Fredholm product)**.** *On*

$$\mathfrak{S} \times \{z \in \mathbb{C} : \operatorname{Re} z > 3\},$$

the family $\mathfrak{L}_{s,z}$ *is holomorphic with values in* $S_1(\mathcal{H}_\oplus)$*. Therefore*

$$\mathfrak{D}(s,z,u) = \det_{\mathcal{H}_\oplus}(I - u\mathfrak{L}_{s,z}) \tag{36}$$

is jointly holomorphic in (s,z,u)*, and*

$$\mathfrak{D}(s,z,u) = \prod_{p \text{ odd}} \mathcal{D}_p(s, up^{-z}). \tag{37}$$

The product is locally normal and independent of the enumeration of the odd primes.

Proof. Fix a compact $K \Subset \mathfrak{S}$ and a compact z-set on which $\operatorname{Re} z \geq 3 + \varepsilon$ for some $\varepsilon > 0$. The trace-norm upper bound from theorem 4.1 gives

$$\sum_{p \text{ odd}} \left\| p^{-z} \mathcal{L}_{s,p} \right\|_{S_1} \leq C_K \sum_{p \text{ odd}} p^{-1-\varepsilon} < \infty, \tag{38}$$

uniformly in s and z on these compact sets. Each block depends holomorphically on s, while $p^{-z} = e^{-z \log p}$ is entire in z. The Weierstrass theorem for Banach-valued holomorphic series makes the direct sum S_1-holomorphic.

The ordinary determinant is continuous and holomorphic on trace class [5]. Finite prime truncations factor into the product of their block determinants. They converge to $\mathfrak{L}_{s,z}$ in trace norm, locally uniformly, so determinant continuity gives equation (37). Absolute trace-norm convergence makes the direct sum, and hence its determinant, invariant under every permutation of the prime blocks. □

5.1 Chronological trace expansion

The global object keeps the exact local chronology rather than collapsing it to a transition statistic.

Proposition 5.2 (Wordwise prime coefficient)**.** *In the domain of theorem 5.1, for every $n \geq 1$,*

$$\operatorname{Tr} \mathfrak{L}_{s,z}^n = \sum_{w \in \Gamma^n} A_{s,w} \sum_{p \text{ odd}} p^{-nz} \Theta_p(g_w). \tag{39}$$

Both sums are absolutely convergent, locally uniformly in (s, z). If a primitive word w of length n is repeated r times, its fibre factor is

$$p^{-nrz} \Theta_p(g_w^r), \tag{40}$$

not $p^{-nrz} \Theta_p(g_w)^r$.

Proof. Powers of a block-diagonal operator remain block diagonal, so

$$\operatorname{Tr} \mathfrak{L}_{s,z}^n = \sum_{p \text{ odd}} p^{-nz} \operatorname{Tr} \mathcal{L}_{s,p}^n.$$

Insert the source-locked word formula equation (18). Since $|\Theta_p(h)| \leq p^2$ and the scalar word trace has a locally uniform absolute majorant, the absolute prime majorant is a constant times

$$\sum_{p \text{ odd}} p^{2-n \operatorname{Re} z},$$

which converges already for $n = 1$ when $\operatorname{Re} z > 3$. Tonelli's theorem therefore permits the stated reordering. The repetition formula follows by applying the actual representation to the chronological matrix power g_w^r. □

For sufficiently small u, the usual expansion

$$\operatorname{Log} \mathfrak{D}(s, z, u) = -\sum_{n \geq 1} \frac{u^n}{n} \operatorname{Tr} \mathfrak{L}_{s,z}^n \tag{41}$$

then inserts equation (39) without changing word order.

5.2 Regularized hierarchy

For an integer $m \geq 2$, corollary 4.3 gives

$$\mathfrak{L}_{s,z} \in S_m \quad \Longleftrightarrow \quad \operatorname{Re} z > \frac{3}{m}. \tag{42}$$

Thus the regularized determinant $\det_m(I - u\mathfrak{L}_{s,z})$ is available in exactly that half-plane; its logarithm near the origin starts at $n = m$. In particular, $z = 2$ gives an operator in $S_2 \setminus S_1$ and first permits $\det_2$, while $z = 1$ first permits $\det_4$. At $z = 0$ the block sum is noncompact, so no finite-order regularized determinant applies. These statements use the standard regularized Fredholm convention [3, 5]. They do not continue the ordinary determinant across its trace-class wall.

The weight p^{-z} is therefore analytically effective and dynamically external. On a length-n return it contributes $e^{-nz \log p}$, adding $z \log p$ to the original AGY roof for each return. Prime damping retains chronology but does not arise from the one-clock dynamical system itself.

6 Normalized-trace collapse on the positive monoid

The direct sum uses counting trace on every p^2-dimensional fibre. A natural alternative is to normalize that trace by p^{-2}. The limit exists, but freeness makes it dynamically trivial on the one-sided positive language.

Theorem 6.1 (Vanishing normalized moments and determinant germ)**.** *For every fixed $n \geq 1$,*

$$\frac{1}{p^2} \operatorname{Tr} \mathcal{L}_{s,p}^n \longrightarrow 0 \tag{43}$$

locally uniformly for $s \in \mathfrak{S}$. More precisely, for every compact $K \Subset \mathfrak{S}$, define

$$M_K = \sup_{s \in K,\ p \text{ odd}} \|\mathcal{L}_{s,p}\| < \infty. \tag{44}$$

Choose any $r_K > 0$ with $r_K < M_K^{-1}$. On $K \times \{|u| \leq r_K\}$, let $\operatorname{Log}_0 \mathcal{D}_p(s,u)$ be the Fredholm logarithm whose value at $u = 0$ is zero. Then, uniformly on this set,

$$\frac{1}{p^2} \operatorname{Log}_0 \mathcal{D}_p(s,u) \longrightarrow 0, \qquad \exp\bigl[p^{-2} \operatorname{Log}_0 \mathcal{D}_p(s,u)\bigr] \longrightarrow 1. \tag{45}$$

Proof. Divide the absolute word trace equation (18) by p^2:

$$\frac{1}{p^2} \operatorname{Tr} \mathcal{L}_{s,p}^n = \sum_{w \in \Gamma^n} A_{s,w} \frac{\Theta_p(g_w)}{p^2}. \tag{46}$$

Every word in this sum is nonempty. Freeness in equation (12) gives $g_w \neq I$, so each normalized character tends to zero by corollary 3.2. Its modulus is at most one. The inherited absolute word majorant permits dominated convergence. Local uniformity follows by the same finite-head/uniform-tail argument used in theorem 4.1: choose a finite word head whose scalar tail is uniformly small on K, and then use pointwise convergence on that finite head. This proves equation (43).

The operator-norm upper bound in theorem 4.1 proves $M_K < \infty$, and the trace-norm branch estimate gives a constant C_K such that

$$\|\mathcal{L}_{s,p}\|_{S_1} \leq C_K p^2. \tag{47}$$

Therefore

$$\frac{1}{p^2}|\operatorname{Tr}\mathcal{L}_{s,p}^n| \le \frac{1}{p^2}\|\mathcal{L}_{s,p}\|_{S_1}\|\mathcal{L}_{s,p}\|^{n-1} \le C_K M_K^{n-1}. \tag{48}$$

Because $r_K M_K < 1$, the Fredholm series

$$\frac{1}{p^2}\operatorname{Log}_0 \mathcal{D}_p(s,u) = -\sum_{n\ge 1}\frac{u^n}{np^2}\operatorname{Tr}\mathcal{L}_{s,p}^n \tag{49}$$

converges on the indicated disc and is dominated, independently of p and s, by the summable series

$$C_K\sum_{n\ge 1}\frac{r_K^n M_K^{n-1}}{n}.$$

Apply equation (43) term by term and then exponentiate. □

Remark 6.2 (Exact source lock)**.** *The character limit is pointwise for each fixed integral matrix. Its use in theorem 6.1 is rigorous for the positive C26 AGY monoid because the fixed-start decoder proves that every nonempty positive word is nonidentity, while the local trace theorem supplies the absolute word majorant. It is not automatically valid for a two-sided language, where nonempty identity-holonomy words may occur. Nor does equation* (45) *assert convergence on the full u-plane or define a product over primes.*

The exponential in equation (45) is the only normalized-root notation used here. A symbol such as $\mathcal{D}_p^{1/p^2}$ would require a global branch choice and is deliberately avoided. Also, normalized trace $p^{-2}\operatorname{Tr}$ is not operator scaling $p^{-2}\mathcal{L}_{s,p}$; after prime summation the latter yields an S_2 but non-trace-class direct sum and a $\det_2$, hence a different object.

7 Orbitwise arithmetic and a fixed-plane MARKED control

The global trace coefficients have arithmetic structure at regular words, but the character generally depends on the orbit. This section separates that theorem from a full-Rauzy fixed-plane stress test.

7.1 Quadratic prime series at regular words

Fix a word w satisfying

$$D_w = \det(I - g_w) \ne 0. \tag{50}$$

Let ε_w be the quadratic Kronecker character determined by the square class of D_w. The regular case of the finite-Weil character formula gives

$$\Theta_p(g_w) = \varepsilon_w(p) \quad \text{for every odd } p \nmid D_w. \tag{51}$$

No quadratic L-data are assigned here when $D_w = 0$.

For a quadratic Dirichlet character χ, define, initially for $\operatorname{Re}\zeta > 1$,

$$P_\chi(\zeta) = \sum_p \chi(p)p^{-\zeta}, \qquad P_\chi^{\mathrm{odd}}(\zeta) = P_\chi(\zeta) - \chi(2)2^{-\zeta}. \tag{52}$$

Then equation (51) gives the exact decomposition

$$\sum_{p \text{ odd}} p^{-\zeta}\Theta_p(g_w) = P^{\text{odd}}_{\varepsilon_w}(\zeta) + E_w(\zeta), \tag{53}$$

where

$$E_w(\zeta) = \sum_{\substack{p \text{ odd} \\ p \mid D_w}} \big(\Theta_p(g_w) - \varepsilon_w(p)\big)p^{-\zeta} \tag{54}$$

is a finite singular-prime correction.

The Euler logarithm gives an optional expression in familiar L-data. For $\operatorname{Re}\zeta > 1$, fix $\operatorname{Log} L(\zeta,\chi)$ by its absolutely convergent Euler series at infinity. Möbius inversion yields

$$P_\chi(\zeta) = \sum_{k\geq 1} \frac{\mu(k)}{k} \operatorname{Log} L(k\zeta, \chi^k). \tag{55}$$

Here χ^k means the pointwise power and may be imprimitive. Formula (55) follows by expanding $\operatorname{Log} L(\zeta,\chi) = \sum_{m\geq 1} P_{\chi^m}(m\zeta)/m$ and applying the divisor identity $\sum_{d\mid n}\mu(d) = 0$ for $n > 1$; all rearrangements are absolutely convergent in the stated half-plane [1]. We make no logarithm choice across an L-zero and claim no continuation of the prime series.

The character ε_w is orbit-dependent. A bounded census in the local finite-Weil stage found 150 distinct arithmetic signatures among 150 sampled induced branches. This is exact finite evidence of fragmentation in that window, not an all-length theorem. Repetition may change D_w, and the correct factor remains $\Theta_p(g_w^r)$, not $\varepsilon_w(p)^r$.

7.2 A general marked convergence threshold

For this subsection, let $J_\star$ be any frozen unimodular integral symplectic form of rank four. Write $\rho_{p,\star}$ for the full finite Weil representation pulled back to that frame and $\Theta_{p,\star}(g) = \operatorname{Tr}\rho_{p,\star}(g \bmod p)$. The proof of lemma 3.1 uses only integral-minor rank stability and the Thomas magnitude formula, so it applies verbatim in this frozen frame.

Proposition 7.1 (Marked absolute threshold)**.** *Fix $g \in \operatorname{Sp}(J_\star,\mathbb{Z})$ and put*

$$k_{\mathbb{Q}} = \dim_{\mathbb{Q}} \ker(g - I).$$

For complex α,

$$\sum_{p \text{ odd}} |p^{-\alpha}\Theta_{p,\star}(g)| < \infty \quad\iff\quad \operatorname{Re}\alpha > 1 + \frac{k_{\mathbb{Q}}}{2}. \tag{56}$$

Proof. By lemma 3.1, outside finitely many primes,

$$|\Theta_{p,\star}(g)| = p^{k_{\mathbb{Q}}/2}.$$

Since $|p^{-\alpha}| = p^{-\operatorname{Re}\alpha}$, the absolute series is, up to finitely many terms, the prime series with exponent $\operatorname{Re}\alpha - k_{\mathbb{Q}}/2$. Its convergence criterion and boundary divergence give equation (56). $\square$

7.3 The C24-P073 full-Rauzy MARKED control

The fixed-plane control comes from the frozen C24 ledger of eventually positive *full labeled Rauzy cycles*. It is not an induced C26 branch. In this paragraph $\Theta_{p,\mathrm{C24}}$ denotes the finite-Weil character pulled back to the frozen C24 symplectic frame. The base-trivialized matrix of cycle P073 is

$$g_{073} = \begin{pmatrix} 1 & 8 & 4 & 4 \\ 1 & 13 & 6 & 6 \\ 1 & 6 & 4 & 3 \\ 1 & 2 & 1 & 2 \end{pmatrix}, \tag{57}$$

and

$$\det(xI - g_{073}) = (x-1)^2(x^2 - 18x + 1). \tag{58}$$

All 3×3 minors of $g_{073} - I$ vanish, while the gcd of its 2×2 minors equals one. Hence $g_{073} - I$ has rank two over every finite field. In the frozen C24 symplectic frame the two-dimensional Thomas quotient form is represented by

$$Q_{073} = \begin{pmatrix} 1 & 0 \\ 8 & -4 \end{pmatrix}. \tag{59}$$

The exact character formula therefore gives, for every odd prime,

$$\Theta_{p,\mathrm{C24}}(g_{073}) = \left(\frac{-4}{p}\right)\left(\frac{-1}{p}\right) p = p. \tag{60}$$

Consequently the dimension-normalized *marked* contribution diverges:

$$\sum_{p \text{ odd}} \frac{\Theta_{p,\mathrm{C24}}(g_{073})}{p^2} = \sum_{p \text{ odd}} \frac{1}{p} = \infty. \tag{61}$$

Among the 146 frozen C24 cycles, the exact fixed-space census is 125, 20, and 1 in dimensions zero, one, and two; P073 is the unique dimension-two entry. Thus equation (61) closes the **full-C24 dimension-normalized MARKED assembly**. It does not show that the positive-prefix C26 induced language contains a fixed-plane word. The all-word fixed-plane gate for that narrower language remains open and is not needed for the sharp Schatten or damped determinant theorems.

8 Interpretation, obstructions, and next architectures

The results give one positive global object and two hard obstructions. Their relationship is summarized by

Assembly	Exact outcome
Unweighted counting-trace direct sum	Noncompact: local operator norms do not tend to zero.
Normalized fibre trace on the positive monoid	The branch-defined determinant germ converges to 1.
Prime-norm damping p^{-z}	Nontrivial ordinary determinant for $\operatorname{Re} z > 3$, with external clock $z \log p$.

This is a canonicality trilemma within the declared architecture. It does not say that every conceivable global construction is impossible. It says that the most direct counting trace is too large, the normalized regular trace sees no nonempty positive holonomy, and the convergent nontrivial alternative requires extra prime grading.

8.1 Why the object is not adelic

An adelic oscillator representation begins with local-field Heisenberg and Weil representations, a global additive character, compatible local splittings and measures, and a restricted tensor product of Schwartz spaces with chosen unramified reference vectors at almost every place; this is the architecture originating in Weil's construction [7]. By contrast, $\mathfrak{L}_{s,z}$ is a Hilbert *direct sum* of representations over the residue fields $\mathbb{F}_p$. There are no p-adic local representations, no adelic Schwartz space, and no product formula tying the local phases together. The term *prime-graded Dirichlet–Fredholm family* is therefore accurate; *adelic Weil representation* is not.

The orbitwise characters in section 7 do not repair this gap. They yield quadratic prime series with word-dependent square classes and finite singular-prime corrections. They do not provide a common conductor, gamma factor, functional equation, or automorphic representation.

8.2 Route-A assessment

The construction clears the operator-definition and determinant-existence gates after damping: the Hilbert space, domain, branch grammar, chronological cocycle, local fibres, sharp trace ideal, and prime-order-independent product are explicit. It also produces exact arithmetic coefficients at regular words. The stronger Hilbert–Pólya gates remain closed. There is no self-adjoint operator, spectral symmetry about a critical line, zero-counting law of Riemann–von Mangoldt type, or divisor identification with a completed L-function. Accordingly, these results authorize a Route-A analytic structure claim but no Route-B or Riemann-hypothesis claim.

8.3 Two large structural pivots

The normalized limit identifies what a genuinely different next step must change.

First, a two-sided based path groupoid would add inverse holonomy. Nonempty reduced loops with identity matrix could then survive the regular trace, unlike words in the free positive monoid. Such a pivot requires a new well-defined trace, control of cancellations, and a fresh nuclearity theorem; it is not obtained by averaging transitions.

Second, a genuine local-field route would replace residue-field fibres by p-adic oscillator representations and assemble them through an adelic restricted tensor product or theta kernel. That route must construct the global additive character and compatible local data before discussing a determinant. Either pivot opens a new architecture rather than adding more finite-prime samples to the current one.

9 Conclusion

Finite-Weil multiplicity is not a removable artifact in the Rauzy–AGY prime fibres. The local Schatten norm has the sharp order $p^{2/q}$, which forces the exact threshold $q \operatorname{Re} z > 3$ for prime-norm damping. Beyond the trace-class wall, the block direct sum produces a jointly holomorphic, prime-order-independent ordinary Fredholm determinant and keeps the full chronological word matrix in every coefficient.

The same theorem also settles the two simplest undamped proposals. Counting trace gives a noncompact direct sum, while normalized trace converges to the regular character and erases the positive-monoid determinant germ. At regular words, arithmetic survives as orbit-dependent quadratic prime series, not as one common automorphic conductor. The full-Rauzy P073 example is an exact dimension-normalized MARKED obstruction in its declared C24 scope and no more.

The next meaningful test is therefore architectural: introduce inverse holonomy through a based groupoid, or build genuine local-field oscillator data. Extending the existing prime table would not address the obstruction proved here.

Data and code availability

The source-lock certificates, independent checker, theorem parameters, and machine-readable outputs are included in the accompanying repository under `henon_dynamics/agy_prime_direct_sum_determinant`. Exact predecessor hashes and verification commands are recorded in appendix A.3.

Ethics statement

This mathematical and computational study uses no human participants, personal data, animals, or field interventions. No ethics approval was required.

Author contributions

The authors jointly contributed to conceptualization, formal analysis, software, validation, visualization, writing—original draft, and writing—review and editing, following the CRediT taxonomy.

Competing interests

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A Proof details and reproducibility passport

A.1 Direct-sum compactness and determinant order

For completeness, suppose T_p is compact on H_p and $T = \bigoplus_p T_p$. If T is compact, choose unit vectors $v_p \in H_p$ with $\|T_p v_p\| \geq \|T_p\| - p^{-1}$. The vectors occupy mutually orthogonal blocks. Compactness of T forces $\|T_p v_p\| \to 0$, hence $\|T_p\| \to 0$. Conversely, if the block norms tend to zero, finite block truncations converge to T in operator norm and are compact. This proves the compactness assertion used in theorem 4.2 without a convention at $q = \infty$.

For trace-class blocks, finite direct sums satisfy

$$\det\left(I - u \bigoplus_{p \leq P} T_p\right) = \prod_{p \leq P} \det(I - uT_p).$$

If $\sum_p \|T_p\|_{S_1} < \infty$, finite block truncations converge in trace norm. Continuity of the Fredholm determinant then proves the infinite product and removes the prime-order choice. This is stronger than conditionally converging a scalar logarithmic Euler product.

For $m \geq 2$, the convention used in equation (42) is

$$\det_m(I - T) = \det\left((I - T)\exp\left(\sum_{j=1}^{m-1} \frac{T^j}{j}\right)\right), \qquad T \in S_m.$$

It deletes the first $m - 1$ trace terms near the origin and does not restore the missing ordinary trace.

A.2 Source-lock dependency graph

The sharp theorem uses only the following implications:

Frozen stage	Imported statement
C25	Theorem E.1: starting labeled state plus full Rauzy matrix decodes the entire edge word. Corollary E.2: based first-return matrices are distinct and their positive concatenations form a free monoid.
C26	Common Bergman domain, locally uniform trace-norm branch sum, bounded constants/evaluation, nowhere-zero slice coefficients, and absolute scalar word trace.
C27	Full p^2-dimensional finite Weil fibre, fixed-prime trace-class determinant, exact Thomas character magnitude, and chronological word trace.
C24	P073 and the 146-cycle full-Rauzy control census only.

In particular, C24-P073 is not used in the proof of theorems 4.1, 4.2 and 5.1. It tests only the full-C24 dimension-normalized MARKED assembly.

A.3 Reproducibility passport

The C28 checker freezes the following SHA-256 source hashes:

Stage	SHA-256
C24	`4b4fe5943262137eeeb3eda4de887725a0663402a1f39f8cc43e089bcc91e778`
C25	`a35cee22714abbb9dc9aadcc165720d1ff77aff3b7f29071f53a1b451760bd12`
C26	`1c0289b9b47e65e0603ea001be7cce263aea13d58c66e4609eac88edf8f7ce4a`
C27	`8676d17c5a0e4444dded88b5a76f5ea1fa974275528aa0a51400730759d8b029`

The executable producer records theorem parameters and exact algebraic sentinels; an independent checker imports no producer module and verifies the hashes before evaluating the certificate. The operator-theoretic lower bound and its finite-head/uniform-tail limit remain proofs, not fields read from a certificate.

To reproduce the computational audit from the project directory, run the documented producer and independent checker commands in the project `README.md`; to rebuild this article, run `latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex` in `paper/`. The compilation report records tool versions, page count, font embedding, unresolved references, and warnings.

A.4 Claim-status firewall

The sharp Schatten bounds, weighted direct-sum equivalence, ordinary determinant for $\operatorname{Re} z > 3$, regular-character limit, and normalized positive-monoid collapse are theorem-level deductions. The P073 identities and 146-cycle census are exact finite controls. The 150-signature conductor census is finite evidence only. Exclusion of fixed planes from every word of the C26 induced language, continuation toward $z = 0$, and any nontrivial groupoid trace remain open.

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